

prettytex
write prettier L^AT_EX

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1 General conventions and ideas

This document is, in theory, only a nice-to-have as all relevant information *should* be written in the corresponding style files. However.

1.1 Introduction

This collection of macros and classes was started during bachelor programmes (Physics and Information and Computer Engineering) for homework assignments and lab-reports. Based on the commit history, we started in 2020, but it is likely that some forms of the collection were individually started earlier.

The goal of *prettytex* is to make writing pretty (thus the name) $\text{\LaTeX}$ documents easy. This is achieved by outsourcing things like package loading, page layout, colours, macros, operators and such to macro-collections grouped into different packages, which are:

- | | | |
|------------------|------------------|-----------------|
| • <i>base</i> | • scriptum | • math-sigtrans |
| • <i>math</i> | • invoice | • math-theorems |
| • gfx | • code | • physics |
| • mathematicians | • contract | • pseudo |
| • shortcuts | • neturalThesis | • thesis |
| • theorems | • gfx-components | • todo |
| • boxes | • gfx-circuits | |

Packages highlighted in **red** are bound to be removed from the main branch, packages marked **orange** are put on the back-burner in the refactor, while a **purple** highlight means, that the package is going to be refactored.

1.2 Package options

We provide some elementary package options, which are handled via *xkeyval*. We have some key-value options, like lang in *base*, but also quite a few boolean-flags, like vargreek in *math*. In order to pass a key-value option, load the package as such:

```
\usepackage[<option>=<value>]{prettytex/<package>}
```

To set a boolean-flag to it's non-default value (every boolean flag has a default which is either **true** or **false**), simply pass the option name as such:

```
\usepackage[<booleanFlag>]{prettytex/<package>}
```

1.3 *prettytex* macros

Each package typically provides a few so called *prettytex-macros*. These are used to keep macros, operators and related things modular to a certain degree. This concept is heavily used in *math*. For example, currently, the `\trace` operator results in trace. However, if you wish to alternatively use, say, `tr`, instead of redefining the `\trace` operator, you can redefine `\prettytexMathTrace` to `tr` and be done with it. *prettytex-macros* follow a very simple syntax:

```
\prettytex<Package><Macro> ,
```

so all *prettytex-macros* in *math* are of the form `\prettytexMath<Macro>`. The exception here is *base*, where it simply is `\prettytex<Macro>`. *prettytex* macros are always listed in their corresponding package description.

To keep everything legible, we will mostly omit when talking about *prettytex-macros*. Whenever this can happen, we highlight it in this documentation.

2 base

base is the entry-point of *prettytex*. It is required by all other packages. It is used to set global options like language, page layout and such and loads elementary packages.

2.1 Package options

option	description	type
lang	the language to use throughout the document (mainly used for babel), expect valid babel language	string
singleLineCheck	the package <i>caption</i> provides the option <code>singlelinecheck</code> which allows to toggle whether or not captions of floats are centered (indented) or not via a boolean flag. false disables centered captions, default is true	boolean
colouredLinks	whether or not to use colouredLinks for reference, hyperrefs etc. default is true . If false , links will instead be highlightet by a coloured underline	boolean
parSkip	the <i>parskip</i> package introduces some small amount of vertical space between paragraphs instead of indenting the new paragraph. If true , the pacakge is loaded. Default is false	boolean
beamer	the <i>beamer</i> package clashes heavily with external page layouts, this if this option is passed, our page layout is disabled. Default is false	boolean

Table 1: Package options for *base*

2.2 Loaded packages

package	description	options
<i>fontenc</i>	T1 font encoding ensures that non-ascii characters like the german umlaute (ö,ä etc) are actual glyphs instead of accented letters. This also improves copy-paste behavior	T1
<i>inputenc</i>	allows for utf8 characters to bes used in source files (e.g. ä.ö. etc)	utf8
<i>lmodern</i>	use latin modern font instead of computer modern	
<i>float</i>	allows for better control of float placement	
<i>geometry</i>	used for page layout (mostly for margins)	
<i>xcolor</i>		table
<i>cellspace</i>		column=C
<i>hyperref</i>	place clickable links in the text (if enabled via options, links are coloured instead of underlined)	hypertextnames=false
<i>translations</i>	allows for automatic translations of text (e.g. theorem names)	
<i>contour</i>	give text an outline of any colour and width	outline
<i>xspace</i>		
<i>subcaption</i>		
<i>graphicx</i>		
<i>parskip</i>	only loaded if enabled via option, see 1 for explanation	
<i>enumitem</i>	more control over enums, specifically their labelling, only loaded if beamer is false	inline

Table 2: Packages loaded by *base*

2.3 Document geometry

The options passed to *geometry* are

- a4paper,
- left=2cm,
- right=2cm,
- top=1.25cm
- bottom=1.25cm
- includeheadfoot

2.4 Colours

prettytex provides a few colours:

color-name	defined as	R	G	B	hex-code
purple	RGB	180	2	212	#b402d4
green	RGB	0	153	51	#009933
red	RGB	204	0	0	#cc0000
blue	RGB	0	102	255	#0066ff
orange	RGB	227	102	0	#e36600
darkred	HTML				#e05263
darkpurple	HTML				#5a016a
pastelpurple	HTML				#a580d1
pastelpink (pastelpink)	HTML				#f8c8dc
lightgrey (lightgrey)	HTML				#dddddd
darkpink	HTML				#e6acac
themecolor	HTML				#00a8ff
themecolor2	HTML				#659157
themecolor3	HTML				#3d518c
themecolor4	HTML				#f25f5c
lightgreybg (lightgreybg)	rgb	0.97	0.97	0.97	
headrulecolor (headrulecolor)	rgb	0.8	0.8	0.8	
sigmared	HTML				#af0f1e
sigmayellow	HTML				#ffd500
discord	HTML				#5865f2

Table 3: Colours defined by *prettytex*

2.5 Formatting

We also provide some formatting macros:

macro	expands to
HighlightColour	pastelpurple
InfoColour	themecolor
WarnColour	orange
ErrorColour	red

Table 4: *prettytex*-macros

macro	arguments	description	example
ebf	text	emphasised and bold-faced text	<i>example</i>
ibf	text	italicised and bold-faced text	<i>example</i>
ibf	text	underlined and bold-faced text	<u>example</u>
eubf	text	emphasised, underlined and bold-faced text	<u><i>example</i></u>
iubf	text	italicised, underlined and bold-faced text	<u><i>example</i></u>
col	colour, text	write text in colour	example
REFL	text	highlight reference, that does not yet exist	example
highlight	text	highlight text	example
info	text	info text	example
warn	text	warn text	example
error	text	error text	example

Table 5: Formatting macros

macro	arguments	description	example/output
<code>cw</code>	<code>text or expression</code>	give the passed argument a white outline with width .06em	example
<code>th</code>		write th as superscript (for dates)	4 th
<code>prettytex</code>		write prettytex in pacakge format	<i>prettytex</i>

Table 6: Miscellaneous macros

2.6 Miscellaneous

macro	arguments	description	example/output
<code>tightlist</code>		remove vertical spacing from all lists	
<code>rom</code>	<code>positive integer</code>	convert input to lowercase roman numeral	xlii
<code>Rom</code>	<code>positive integer</code>	convert input to uppercase roman numeral	LXIX
<code>alp</code>	<code>integer in [1,26]</code>	convert input to lowercase alph-value	x
<code>Alp</code>	<code>integer in [1,26]</code>	convert input to uppercase alph-value	P
<code>listfigures</code>			
<code>listtables</code>			
<code>autoheadings</code>			

Table 7: Miscellaneous macros from *base*

3 *math*